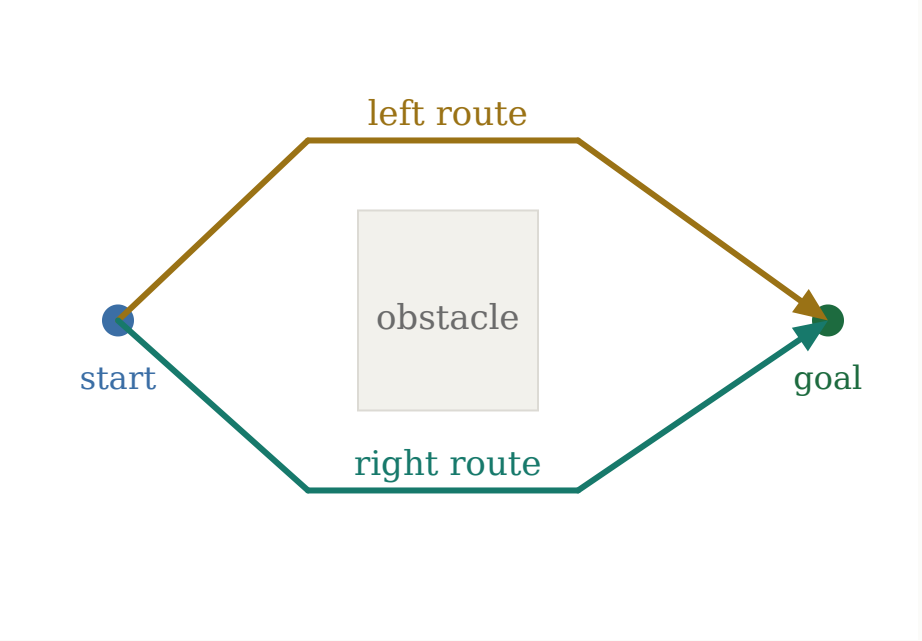
MOTIVATION

Information Theory, Sampling and Density

How should a robot compare uncertain predictions?

Two safe routes can have different probabilities. A useful model must represent both, and we need evidence that it does.



Lecture 9 · November 11, 2026 · Mathematics for Robotics · ROB-GY 6013
NYU Tandon · Jing Zhang · Fall 2026

Goodfellow et al., Deep Learning (2016), §3.13

ROADMAP

The reasoning path

One robot problem, several different questions

We will compare route predictions, measure sensor information, assess trial evidence, locate action groups, and compute probabilities reliably.

Questions for the same robot

1. Compare route predictions
2. Measure sensor information
3. Assess evidence from trials
4. Locate action groups
5. Compute probabilities reliably

Goodfellow et al., Deep Learning (2016), §3.13

RECALL

Probability recall

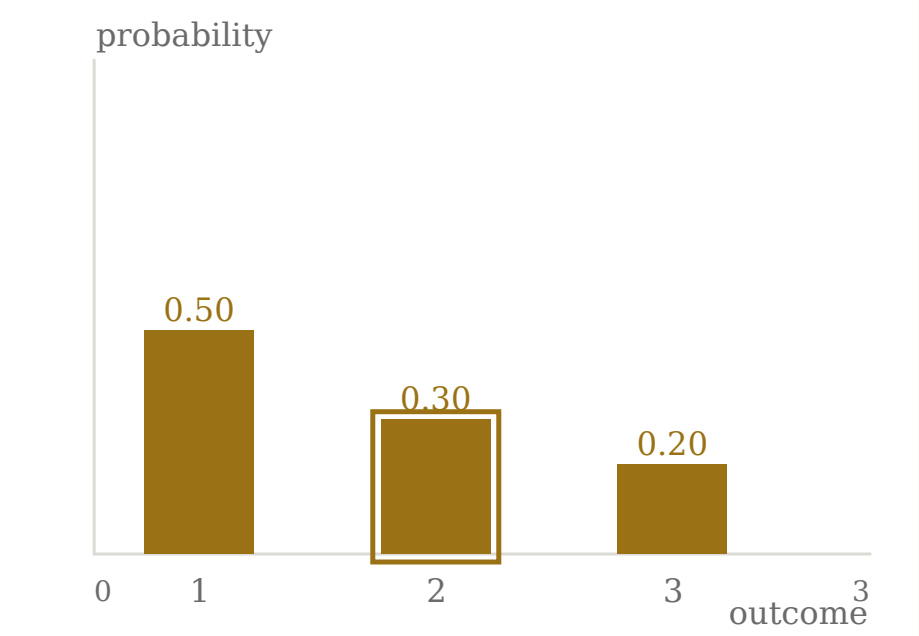
The distribution and its samples are different objects

For a finite outcome set, expectation is a probability-weighted average. Here p is fixed; sampled outcomes vary.

$$\mathbb{E}_p[g(X)] = \sum_x p(x)g(x)$$

`X,x` random outcome and possible value

`p,g` probability mass function and measured quantity



Goodfellow et al., Deep Learning (2016), §3.8; original example

DEFINITION

Surprisal

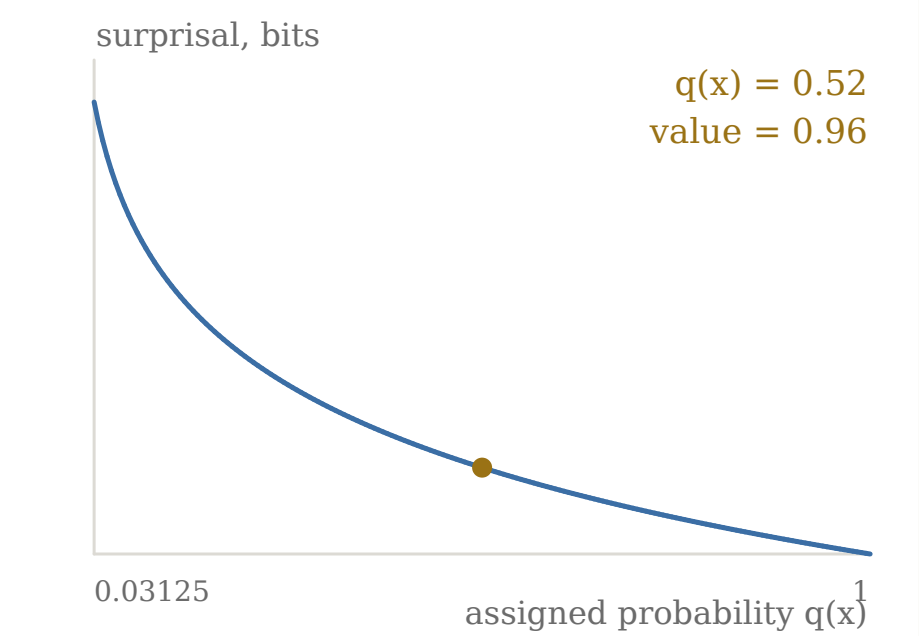
A confident mistake should cost more

Surprisal is the negative logarithm of the probability assigned to an observed event. Here $0<q(x)\leq 1$ and the log base is 2.

$$I_q(x) = -\log_2 q(x)$$

`q(x)` predicted probability of the observed event

`I_q(x)` surprisal, measured in bits



Goodfellow et al., Deep Learning (2016), §3.13

DERIVATION

Surprisal

Independent evidence adds in log space

For independent events, their joint probability is the product.

$$-\log_2(q_1 q_2) = -\log_2 q_1 - \log_2 q_2$$

`q_1,q_2` probabilities of independent events

Goodfellow et al., Deep Learning (2016), §3.13

Surprisal

Independent evidence adds in log space

Factor the independent joint probability

q_1 and q_2 are positive probabilities of independent events.

1. $q(x_1, x_2) = q_1 q_2$ by independence.
2. $I_q(x_1, x_2) = -\log_2(q_1 q_2)$ by the definition of surprise.
3. $-\log_2(q_1 q_2) = -\log_2 q_1 - \log_2 q_2$ by the log-product rule.

Goodfellow et al., Deep Learning (2016), §3.13

DEFINITION

Entropy

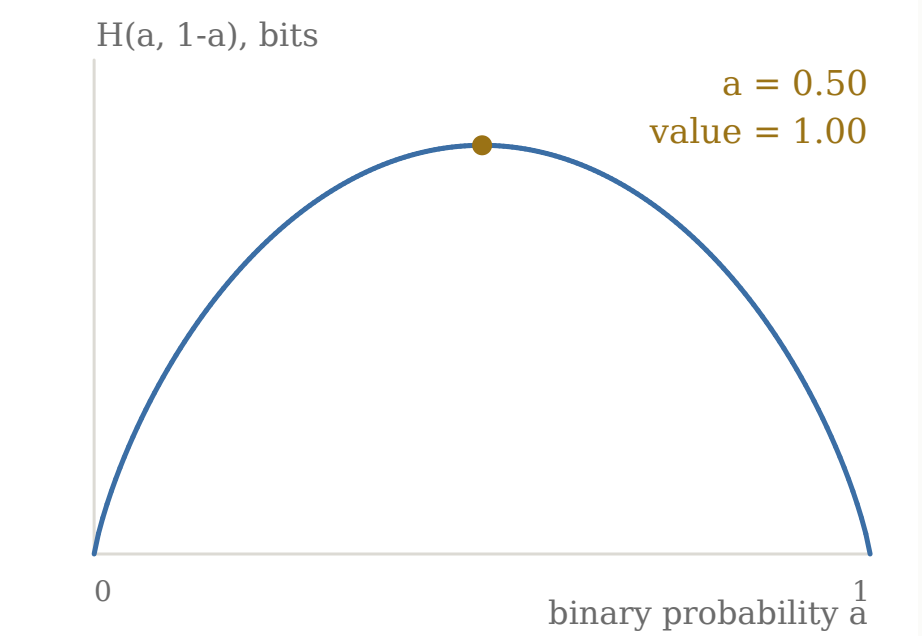
Average surprise under the true distribution

Entropy is the expected surprisal of an outcome under its own distribution. The curve shows binary distributions (a,1−a).

$$H(p) = -\sum_{x} p(x) \log_2 p(x)$$

`p(x)` probability of outcome x

`H(p)` entropy in bits; use 0 log 0 = 0



Goodfellow et al., Deep Learning (2016), §3.13

EXAMPLE

Entropy

A concrete prefix code makes the units visible

COURSE-CREATED WORKED EXAMPLE

A prefix code has no codeword that begins another. For probabilities (1/2,1/4,1/8,1/8), use (0,10,110,111).

$$L = \tfrac{1}{2}(1) + \tfrac{1}{4}(2) + \tfrac{1}{8}(3) + \tfrac{1}{8}(3)$$

L expected codeword length in bits

symbol	probability	codeword
A	1/2	0
B	1/4	10
C	1/8	110
D	1/8	111

Shannon (1948), §§6,9–12; original course examples

EXAMPLE

Entropy

The code length equals entropy in this example

COURSE-CREATED WORKED EXAMPLE

The same four probabilities give ideal lengths 1, 2, 3, 3.

$$H(p) = \tfrac{1}{2}\log_2 2 + \tfrac{1}{4}\log_2 4 + \tfrac{1}{4}\log_2 8 = 1.75$$

`p` (1/2, 1/4, 1/8, 1/8)

`H(p)` average surprise in bits

symbol	probability	codeword
A	1/2	0
B	1/4	10
C	1/8	110
D	1/8	111

Shannon (1948), §§6,9–12; original course examples

EXERCISE

Entropy

Which binary source is hardest to predict?

Compare A: p=(1,0), B: p=(3/4,1/4), C: p=(1/2,1/2).

$$H(p) = -\sum_{x} p(x) \log_2 p(x)$$

`H` entropy in bits; 0 log 0 = 0

- (A) A has the largest entropy
- (B) B has the largest entropy
- (C) C has the largest entropy

Goodfellow et al., Deep Learning (2016), §3.13

DEFINITION

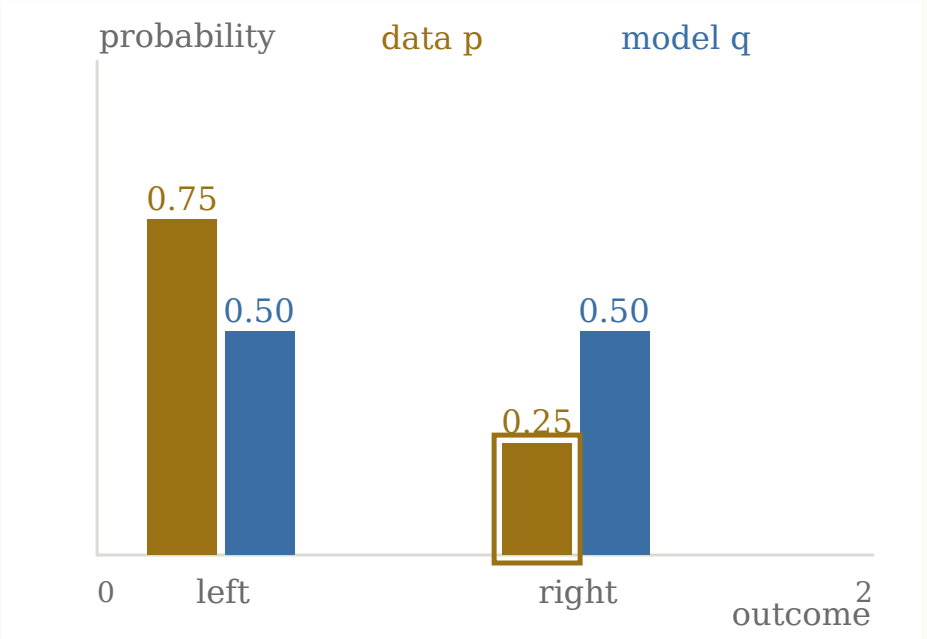
Cross entropy

The source and the prediction can differ

Cross entropy is the expected surprisal under model q when outcomes follow source p. Require q(x)>0 wherever p(x)>0.

$$H(p, q) = -\sum_{x} p(x) \log_2 q(x)$$

- `p` data-generating probabilities
- `q` model probabilities
- `H(p,q)` cross entropy in bits



Goodfellow et al., Deep Learning (2016), §3.13

DEFINITION

KL divergence

The excess cost of using the wrong probabilities

Kullback–Leibler (KL) divergence is the expected log ratio of source p to model q. Use q(x)>0 wherever p(x)>0.

$$D_{\mathrm{KL}}(p\|q) = \sum_{x} p(x) \log_2 \frac{p(x)}{q(x)}$$

`p,q` source and model on the same outcomes

`D_KL` Kullback–Leibler divergence in bits

Goodfellow et al., Deep Learning (2016), §3.13

DERIVATION

KL divergence

The cross-entropy identity follows by expansion

Separate the logarithm of a ratio into a difference.

$$H(p, q) = H(p) + D_{\mathrm{KL}}(p \| q)$$

`H(p)` source uncertainty

`D_KL` extra cost of q

`H(p,q)` total average log penalty

Goodfellow et al., Deep Learning (2016), §3.13

KL divergence

The cross-entropy identity follows by expansion

Expand the log ratio

p is the fixed source; q is positive wherever p is positive.

1. $D_{\mathrm{KL}}(p\|q) = \sum_x p(x) \log_2 [p(x)/q(x)]$.
2. $= \sum_x p(x) \log_2 p(x) - \sum_x p(x) \log_2 q(x)$ by the log-ratio rule.

Goodfellow et al., Deep Learning (2016), §3.13

KL divergence

The cross-entropy identity follows by expansion

Identify the two entropy terms

H(p) is source entropy; H(p,q) is cross entropy.

1. $\sum_x p(x) \log_2 p(x) = -H(p)$.
2. $-\sum_x p(x) \log_2 q(x) = H(p, q)$.
3. Hence $D_{\mathrm{KL}}(p \| q) = H(p, q) - H(p)$, or $H(p, q) = H(p) + D_{\mathrm{KL}}(p \| q)$.

Goodfellow et al., Deep Learning (2016), §3.13

EXAMPLE

KL divergence

A two-route model has a measurable penalty

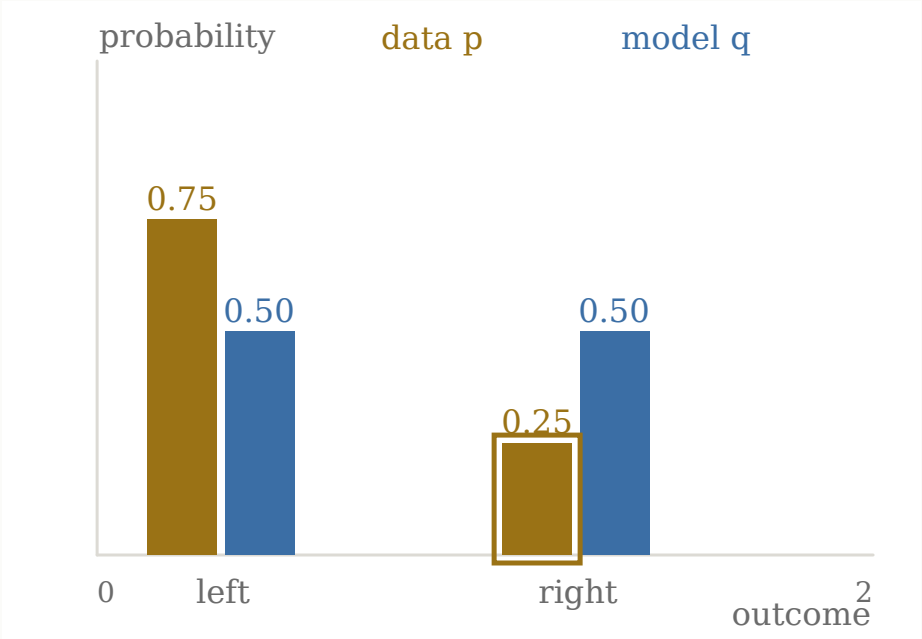
COURSE-CREATED WORKED EXAMPLE

Source p=(3/4,1/4). Model q=(1/2,1/2).

$$H(p,q) = 1, \quad H(p) \approx 0.811, \quad D_{\mathrm{KL}} \approx 0.189$$

`p,q` route probabilities in left/right order

`values` bits per observed route



[Goodfellow et al., Deep Learning (2016), §3.13]

EXERCISE

KL divergence

What changes when source and model swap?

Now p=(1/2,1/2), q=(3/4,1/4). Find KL in bits.

$$D_{\mathrm{KL}}(p\|q) = \sum_{x} p(x)\log_2[p(x)/q(x)]$$

`p,q` source and model on the same two outcomes

A About 0.189

B About 0.208

C Exactly zero

Goodfellow et al., Deep Learning (2016), §3.13

DERIVATION

Cross entropy

Maximum likelihood uses the same average penalty

For a fixed dataset, maximum likelihood chooses parameters that maximize its model probability. Assume iid discrete observations from qθ.

$$-\frac{1}{n}\log\prod_{i=1}^{n} q_\theta(x_i) = -\frac{1}{n}\sum_{i=1}^{n}\log q_\theta(x_i)$$

`θ` adjustable model parameters

`n,x_i` fixed sample count and observed values

Goodfellow et al., Deep Learning (2016), §5.5

Cross entropy

Maximum likelihood uses the same average penalty

Turn a likelihood product into a sum

$x_1,\ldots,x_n$ are observed iid data; θ is adjustable; log is natural.

1. $L(\theta) = \prod_{i=1}^{n} q_\theta(x_i)$ by the iid model.
2. $-n^{-1} \log L(\theta) = -n^{-1} \sum_i \log q_\theta(x_i)$ by the log-product rule.

Goodfellow et al., Deep Learning (2016), §5.5

Cross entropy

Maximum likelihood uses the same average penalty

Group repeated outcomes

N_x is the count of outcome x; empirical frequency is p̂ₙ(x)=N_x/n.

1. $-n^{-1}\sum_i \log q_\theta(x_i) = -\sum_x (N_x/n)\log q_\theta(x)$.
2. $= -\sum_x \hat{p}_n(x)\log q_\theta(x)$, the empirical cross entropy.
3. The minus sign converts maximization to minimization; fixed 1/n does not change the optimizer.

Goodfellow et al., Deep Learning (2016), §5.5

EXERCISE

KL divergence

Can a model forbid an event that occurs?

p=(0.9,0.1), q=(1,0). The second outcome is possible under p.

A KL(p||q) is infinite

B KL(p||q) is zero

C The zero-probability term is ignored

Goodfellow et al., Deep Learning (2016), §3.13

DERIVATION

KL divergence

Why the total cannot be negative

Use natural logs for this proof: $-\ln u \geq 1-u$ for $u>0$. The resulting KL is in nats; changing log base preserves its sign.

$$D_{\mathrm{KL}}(p\|q) \geq \sum_{p(x)>0} [p(x) - q(x)] \geq 0$$

- `u` q(x)/p(x)
- `log` natural logarithm here; base changes only scale

Goodfellow et al., Deep Learning (2016), §3.13

KL divergence

Why the total cannot be negative

Apply the tangent inequality

Use natural logs; u=q(x)/p(x)>0 on the support of p.

1. $-\log u \geq 1 - u$, with equality only at u=1.
2. $p(x)\log[p(x)/q(x)] \geq p(x) - q(x)$ after multiplying by p(x)>0.

Goodfellow et al., Deep Learning (2016), §3.13

KL divergence

Why the total cannot be negative

Sum over source support

S={x:p(x)>0}; both p and q have total mass 1.

1. $D_{\mathrm{KL}}(p\|q) \geq \sum_{x\in S}(p(x)-q(x))$.
2. $= 1 - \sum_{x\in S} q(x) \geq 0$.
3. Equality requires q=p on S and no q mass outside S. A log-base change preserves the sign.

Goodfellow et al., Deep Learning (2016), §3.13

COUNTEREXAMPLE

KL divergence

Nonnegative does not make it a distance

COURSE-CREATED WORKED EXAMPLE

KL can be asymmetric and violate the triangle inequality.

$$D(p\|r) \approx 1.758$$

$$D(p\|q) + D(q\|r) \approx 0.368 + 0.511 < 1.758$$

`p,q,r` (0.9,0.1), (0.5,0.5), (0.1,0.9)

`D` KL in nats for this example

Goodfellow et al., Deep Learning (2016), §3.13

INTERPRETATION

KL fitting

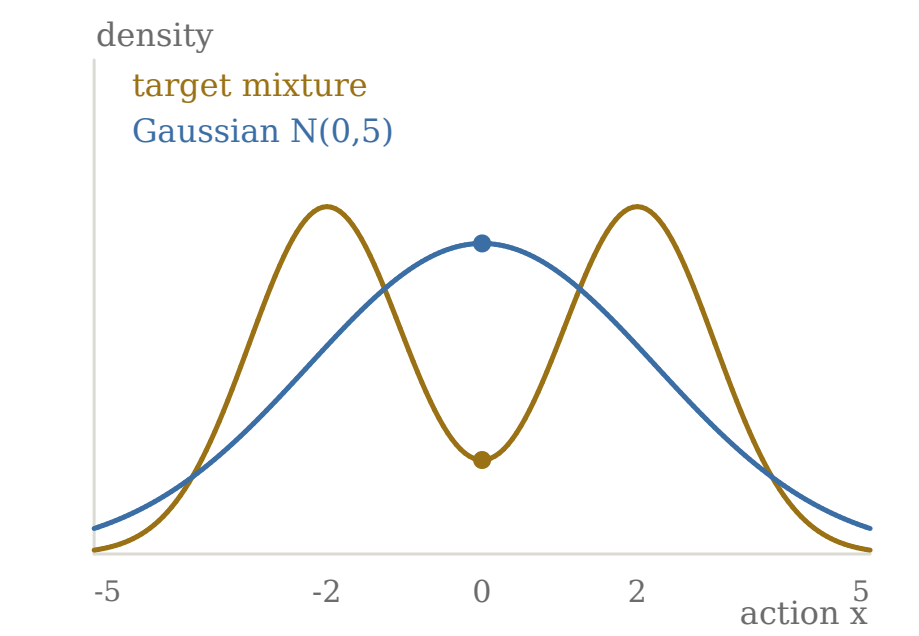
The direction changes what the model averages

A mode maximizes probability mass or density. Fix target p and restrict adjustable qθ to a chosen model family.

$$\min_{\theta} D(p \| q_\theta) \quad \text{or} \quad \min_{\theta} D(q_\theta \| p)$$

`θ` adjustable model parameters

`p,q_theta` target distribution and approximation



Goodfellow et al., Deep Learning (2016), §3.13

DERIVATION

KL fitting

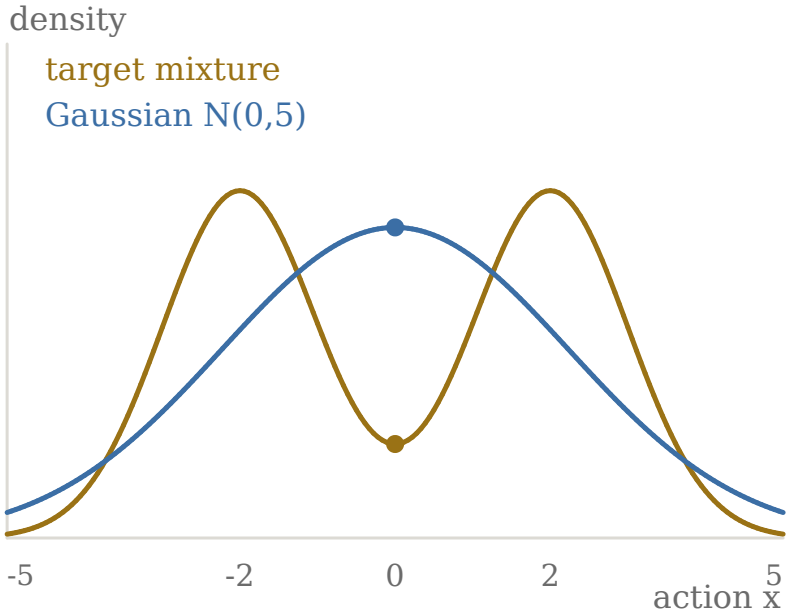
Forward KL within a Gaussian family

Continuous density p on R: finite second moment, finite differential entropy, variance v>0. Fit q=N(m,s²), s²>0.

$$-\mathbb{E}_p[\log q(X)] = \tfrac{1}{2}\log(2\pi s^2) + \frac{\mathbb{E}_p[(X-m)^2]}{2s^2}$$

`m,s²` adjustable Gaussian mean and variance

`mu,v` fixed target mean and variance



Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

KL fitting

Forward KL within a Gaussian family

Insert the Gaussian model density

p is the fixed target density. q=N(m,s²), with adjustable m and s²>0.

1. $q(x) = (2\pi s^2)^{-1/2} \exp[-(x-m)^2/(2s^2)]$.
2. $-\log q(x) = \frac{1}{2}\log(2\pi s^2) + (x-m)^2/(2s^2)$.

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

KL fitting

Forward KL within a Gaussian family

Average under the continuous target

$\mu=\mathrm{E}_p[X]$, $v=\mathrm{Var}_p(X)>0$; differential entropy $h(p)=-\int p(x)\log p(x)\,dx$ is finite.

1. $C(m, s^2) = -\mathbb{E}_p[\log q(X)] = \frac{1}{2}\log(2\pi s^2) + \mathbb{E}_p[(X-m)^2]/(2s^2).$
2. $D_{\mathrm{KL}}(p\|q) = \int p(x)\log[p(x)/q(x)]\,dx = C(m, s^2) - h(p).$
3. Finite fixed h(p) makes minimizing KL equivalent to minimizing C.

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

DERIVATION

KL fitting

The best Gaussian center is the target mean

Target mean $\mu = E_p[X]$, variance $v = E_p[(X-\mu)^2]$. Adjustable m is the Gaussian mean; its variance $s^2 > 0$ stays fixed.

$$\mathbb{E}_p[(X-m)^2] = v + (\mu - m)^2$$

- v target variance $E[(X-\mu)^2]$
- m candidate Gaussian mean

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

KL fitting

The best Gaussian center is the target mean

Expand the square around the target mean

$\mu = E_p[X]$, $v = \mathrm{Var}_p(X)$, m is the adjustable Gaussian mean.

1. $X - m = (X - \mu) + (\mu - m).$
2. $(X - m)^2 = (X - \mu)^2 + 2(X - \mu)(\mu - m) + (\mu - m)^2.$

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

KL fitting

The best Gaussian center is the target mean

Average and minimize

μ−m is fixed when taking expectation over X.

1. $\mathbb{E}[X-\mu]=0$, so the cross term vanishes.
2. $\mathbb{E}[(X-m)^2]=v+(\mu-m)^2$.
3. For fixed $s^2>0$, the nonnegative square is minimized at m=μ.

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

DERIVATION

KL fitting

The best Gaussian variance matches the spread

Fix m=μ. Target variance v>0 is fixed. Adjustable variance r=s²>0 minimizes the expected negative log density C(r).

$$C(r) = \tfrac{1}{2}\log(2\pi r) + \frac{v}{2r}, \qquad C\prime(r) = \frac{r - v}{2r^2}$$

- r candidate Gaussian variance
- v positive target variance

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

KL fitting

The best Gaussian variance matches the spread

Differentiate with respect to the variance

r=s²>0 is adjustable; v>0 is the target variance.

1. $C(r) = \frac{1}{2}\log(2\pi r) + v/(2r).$
2. $C\prime(r) = 1/(2r) - v/(2r^2) = (r - v)/(2r^2).$

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

KL fitting

The best Gaussian variance matches the spread

Verify the minimum, not just stationarity

The denominator $2r^2$ is positive for $r>0$.

1. For $r<v$, $C'(r)<0$. For $r>v$, $C'(r)>0$.
2. C decreases toward v and increases after v, so $r=v$ is the global minimum.
3. The optimal Gaussian has mean μ and variance v.

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

EXAMPLE

KL fitting

One Gaussian can place its mean between two peaks

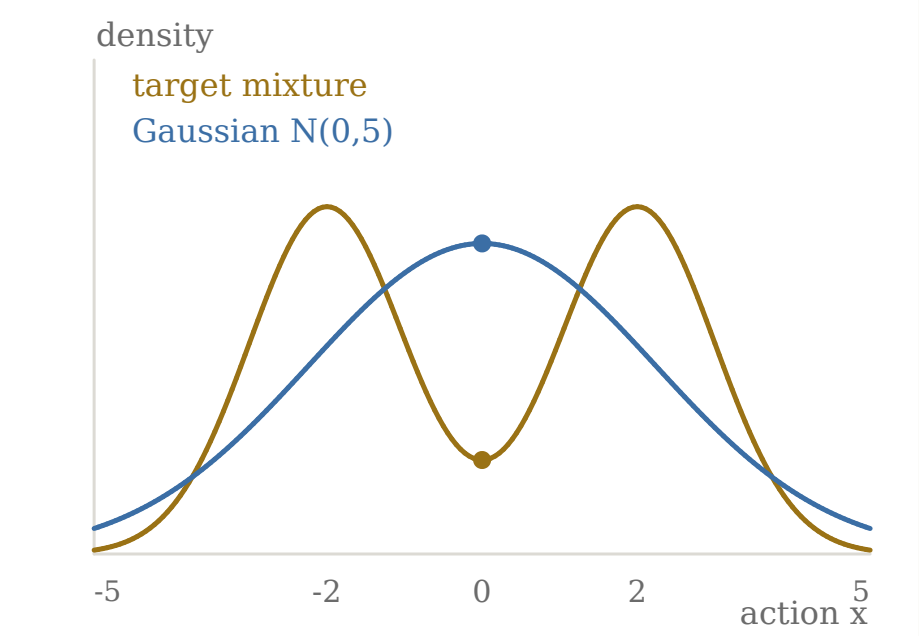
COURSE-CREATED WORKED EXAMPLE

A mixture first chooses a component, then samples it. Here each component $\mathrm{N}(-2,1)$ or $\mathrm{N}(2,1)$ has weight 1/2 and variance 1.

$$\mu = 0, \qquad v = 1 + 4 = 5, \qquad q^* = \mathcal{N}(0, 5)$$

p two-component target density

q* best single Gaussian for forward KL



Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

EXERCISE

KL fitting

An uneven mixture shifts both fitted moments

p=0.75 N(−2,1)+0.25 N(2,1). Fit one Gaussian by forward KL.

$$m = \mathbb{E}_p[X], \qquad s^2 = \mathbb{E}_p[X^2] - m^2$$

`m,s²` forward-KL Gaussian mean and variance

A. Mean −1, variance 1

B. Mean −1, variance 4

C. Mean 0, variance 5

Goodfellow et al., Deep Learning (2016), §§3.8,3.9.3,3.13; course derivation

INTERPRETATION

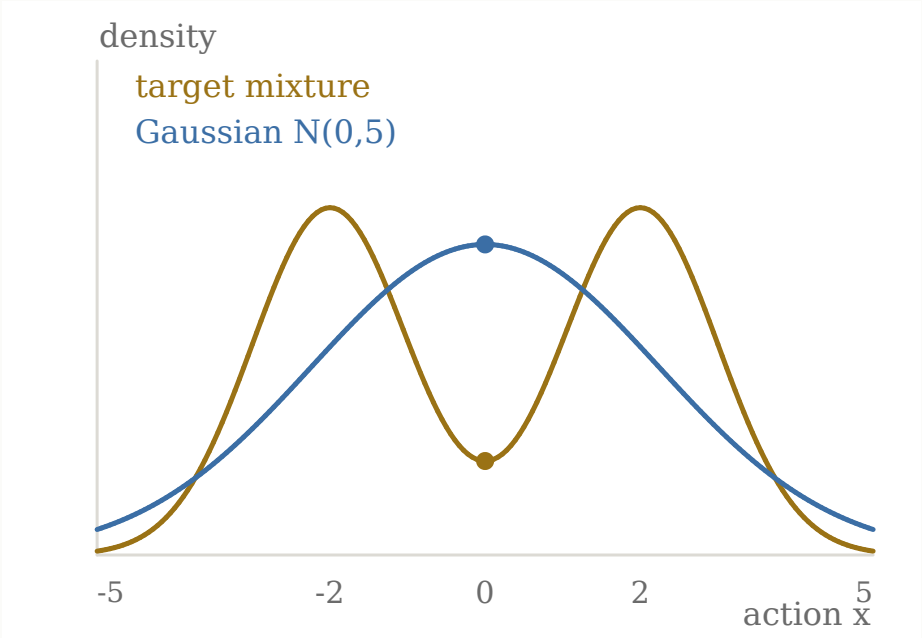
KL fitting

Reverse KL penalizes probability in a forbidden gap

If q(x)>0 where p(x)=0 on a region of positive q mass, reverse KL is infinite.

$$D(q\|p) = \mathbb{E}_q[\log q(X) - \log p(X)]$$

- p fixed target density
- q candidate model density



Goodfellow et al., Deep Learning (2016), §3.13

EXERCISE

KL fitting

Does reverse KL always select one mode?

The candidate family contains the exact bimodal target p.

A Yes, one mode is always optimal

B No, q=p minimizes both directions

C Neither direction can reach zero

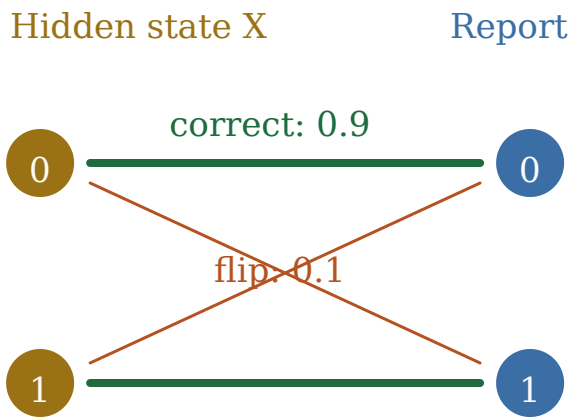
Goodfellow et al., Deep Learning (2016), §3.13

MOTIVATION

Mutual information

A sensor is useful when it changes our uncertainty

Let X be a hidden state and Y a sensor report. Does knowing Y help us predict X?



$P(X=0) = P(X=1) = 0.5$

Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

EXAMPLE

Mutual information

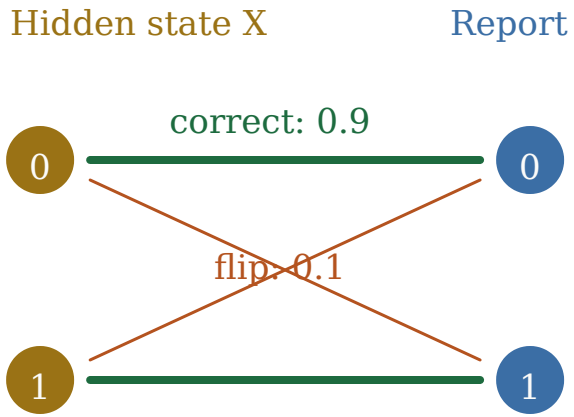
The marginals come from summing the joint table

COURSE-CREATED WORKED EXAMPLE

A fair binary state is reported correctly with probability 0.9.

$$p_{XY} = \begin{bmatrix} 0.45 & 0.05 \\ 0.05 & 0.45 \end{bmatrix}$$

`rows`	X=0,1
`columns`	Y=0,1
`p_X,p_Y`	row and column sums, each (1/2,1/2)



P(X=0) = P(X=1) = 0.5

Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

DEFINITION

Mutual information

Compare dependence with the independent baseline

Mutual information measures dependence as KL from a joint distribution to the product of its marginals. Here X,Y are discrete.

$$I(X;Y) = \sum_{x,y} p(x,y) \log_2 \frac{p(x,y)}{p_X(x) p_Y(y)}$$

`I(X;Y)` mutual information in bits

`p_X,p_Y` marginal distributions

Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

DERIVATION

Mutual information

Information is expected uncertainty reduction

Conditional entropy H(X|Y) is the average entropy of X after observing Y. Factor p(x,y)=pY(y)p(x|y).

$$I(X;Y) = H(X) - H(X \mid Y)$$

`H(X|Y)` average entropy of X after observing Y

`H(X)` entropy before observing Y

Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

Mutual information

Information is expected uncertainty reduction

Cancel the report probability

p(x,y)=p_Y(y)p(x|y) on positive joint-probability pairs.

1. $\log_2 \frac{p(x,y)}{p_X(x)p_Y(y)} = \log_2 \frac{p(x|y)}{p_X(x)}.$
2. $= \log_2 p(x \mid y) - \log_2 p_X(x).$

Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

Mutual information

Information is expected uncertainty reduction

Average over the joint distribution

H(X|Y)=−Σₓ,ᵧp(x,y)log₂p(x|y) is conditional entropy.

1. $I(X;Y) = \sum_{x,y} p(x,y) \log_2 p(x \mid y) - \sum_{x,y} p(x,y) \log_2 p_X(x)$.
2. The first sum is −H(X|Y). Summing over y makes the second sum −H(X).
3. Therefore I(X;Y)=H(X)−H(X|Y), an expected reduction.

Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

EXAMPLE

Mutual information

A noisy binary sensor supplies part of a bit

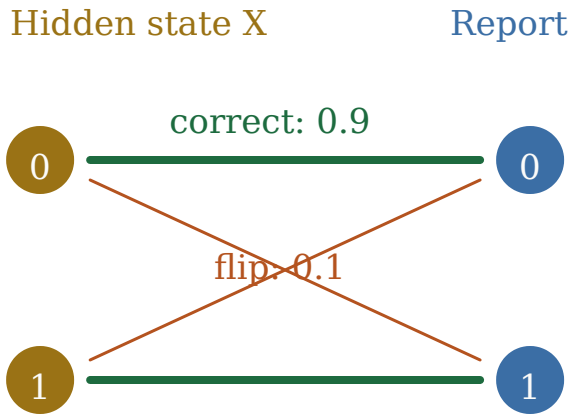
COURSE-CREATED WORKED EXAMPLE

X is fair; Y flips it independently with probability e=0.1. Binary entropy $h_2(e)$ is the entropy of probabilities (e,1−e).

$$I(X;Y) = 1 - h_2(e) \approx 0.531$$

`h_2(e)` $-e\log_2 e-(1-e)\log_2(1-e)$

`e` independent binary flip probability



P(X=0) = P(X=1) = 0.5

Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

EXERCISE

Mutual information

A noisier sensor leaves more uncertainty

X is fair. Y flips X independently with probability e=0.25. Find I(X;Y).

$$I(X;Y) = H(X) - H(X \mid Y)$$

`H(X|Y)` average remaining binary entropy

A. About 0.811 bits

B. About 0.189 bits

C. Exactly 0.75 bits

Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

EXAMPLE

Mutual information

Zero covariance can hide a nonlinear dependence

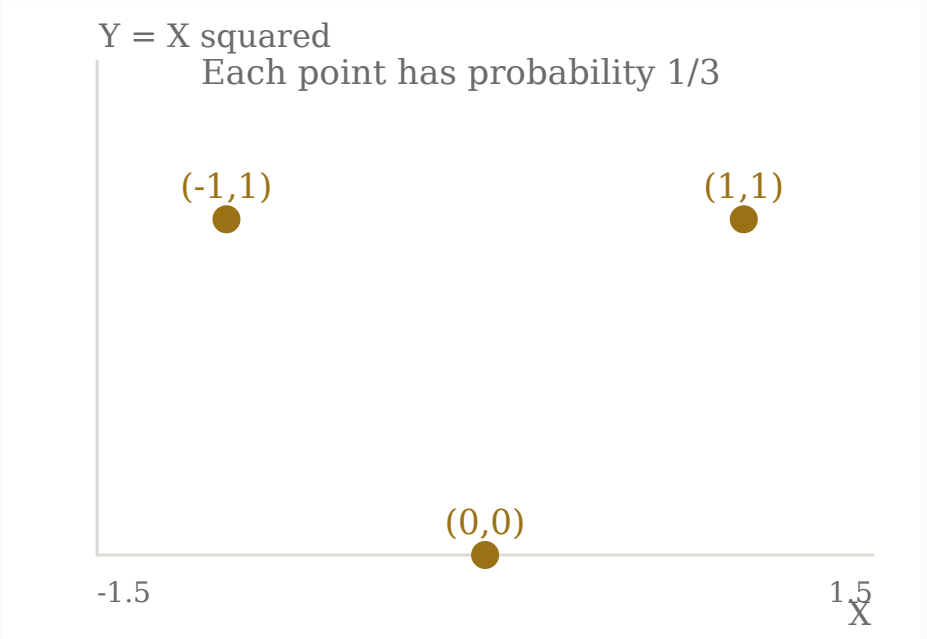
COURSE-CREATED WORKED EXAMPLE

X is uniform on $\{-1,0,1\}$. Set $Y=X^2$.

$$\operatorname{Cov}(X,Y) = \mathbb{E}[X^3] - \mathbb{E}[X]\mathbb{E}[X^2] = 0$$

`X,Y` random input and its square

`Cov` covariance; both variances are positive



[Goodfellow et al., Deep Learning (2016), §3.8; original example]

EXERCISE

Mutual information

How much does the square reveal?

X is uniform on $\{-1,0,1\}$; $Y=X^2$. $H(Y|X)=0$.

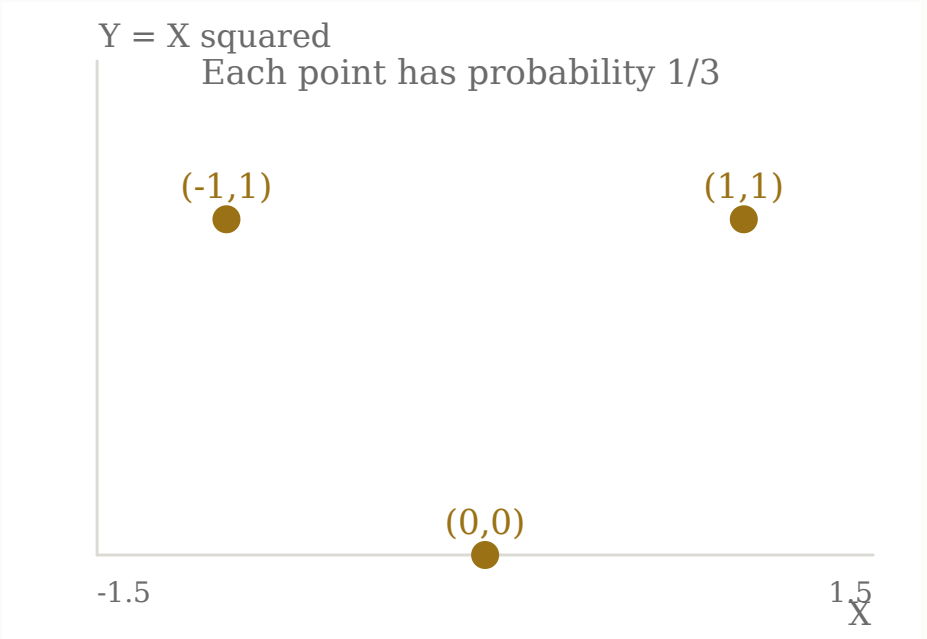
$$I(X;Y) = H(Y) - H(Y \mid X)$$

`H(Y)` entropy of the squared output

A. I(X;Y)=0 because covariance is zero

B. $I(X;Y)=\log_2 3$

C. $I(X;Y)=h_2(1/3)\approx 0.918$ bits



Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

EXERCISE

Mutual information

Compare a direct and an inverted sensor

X is a fair binary state. Sensor A gives $Y=X$. Sensor B gives $Y=1-X$.

A. A has more mutual information

B. Both provide 1 bit

C. B provides zero information

Weissman, Stanford EE376A (2018), Lecture3, Definitions5–6, eqs.(30)–(46)

RECAP

From distributions to data

The formulas need evidence from observations

Entropy measures source uncertainty. KL measures model mismatch. Mutual information measures dependence. Data let us estimate expectations.

Questions for the same robot

1. Compare route predictions
2. Measure sensor information
3. Assess evidence from trials
4. Locate action groups
5. Compute probabilities reliably

Goodfellow et al., Deep Learning (2016), §3.13

DEFINITION

Monte Carlo estimation

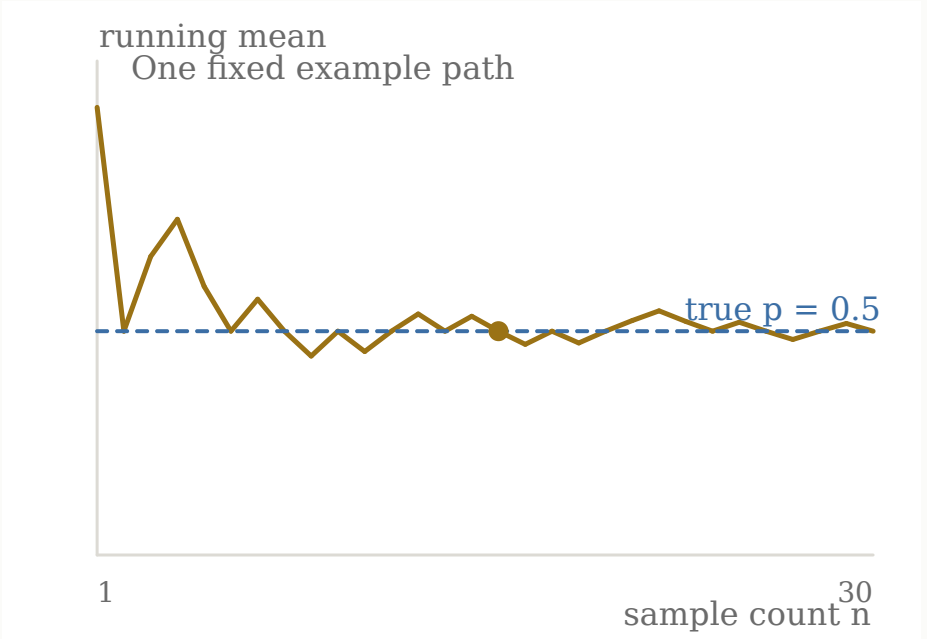
An average estimates a chosen expectation

Monte Carlo estimation approximates an expectation by averaging sampled values. Here n is fixed and draws from p are iid.

$$\hat{\mu}_n = \frac{1}{n}\sum_{i=1}^{n} g(X_i), \qquad \mu = \mathbb{E}_p[g(X)]$$

`X_i` independent draws from the same distribution

`g,mu` measured quantity and its true mean



UC Berkeley CS70 (2017), Note18: mean, variance, concentration

DERIVATION

Monte Carlo estimation

Linearity explains unbiasedness

An estimator is unbiased if its expectation equals its target. Here n is fixed and each g(Xᵢ) has finite expectation μ.

$$\mathbb{E}[\hat{\mu}_n] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[g(X_i)] = \mu$$

μˆ random sample average

μ common expectation of each sample value

UC Berkeley CS70 (2017), Note18: mean, variance, concentration

Monte Carlo estimation

Linearity explains unbiasedness

Use linearity of expectation

n is fixed; every measured value g(Xᵢ) has the same finite expectation μ.

1. $\mathbb{E}[\hat{\mu}_n] = \mathbb{E}[n^{-1} \sum_i g(X_i)]$.
2. $= n^{-1} \sum_i \mathbb{E}[g(X_i)]$ by linearity.
3. $= n^{-1}(n\mu) = \mu$ by the common sample expectation.

[UC Berkeley CS70 (2017), Note18: mean, variance, concentration]

DERIVATION

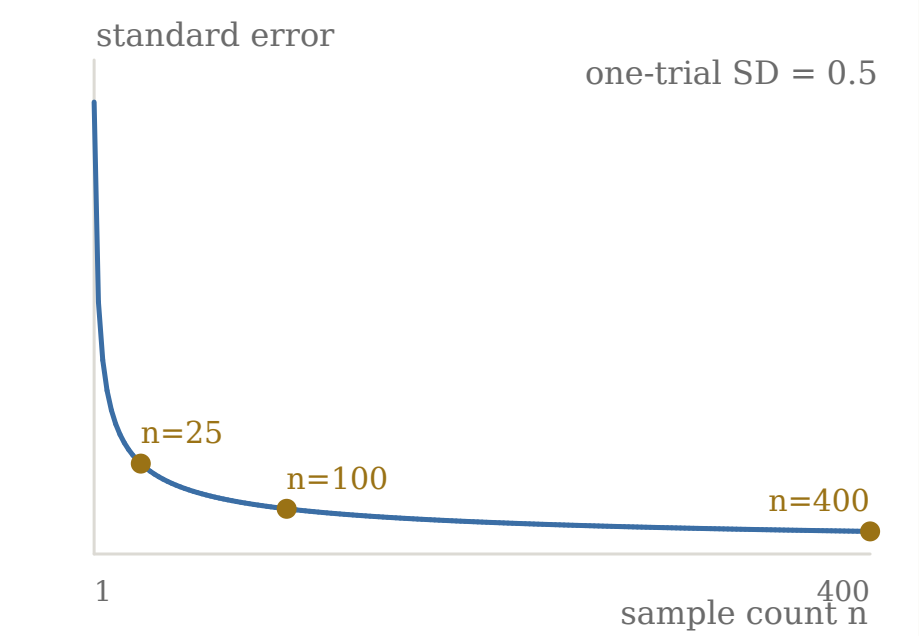
Monte Carlo estimation

Independence makes the variance shrink

Standard error (SE) is an estimator’s standard deviation. Use n fixed independent measurements with common finite variance σ².

$$\mathrm{Var}(\hat{\mu}_n) = \frac{\sigma^2}{n}, \qquad \mathrm{SE}(\hat{\mu}_n) = \frac{\sigma}{\sqrt{n}}$$

- σ² variance of one measured value
- SE standard deviation of the estimator



[UC Berkeley CS70 (2017), Note18: mean, variance, concentration]

Monte Carlo estimation

Independence makes the variance shrink

Expand the variance of the average

$Z_i = g(X_i)$ are independent with common finite variance σ^2.

1. $\operatorname{Var}(\hat{\mu}_n) = n^{-2} \operatorname{Var}(\sum_i Z_i)$.
2. $= n^{-2} \sum_i \operatorname{Var}(Z_i)$ because all cross covariances vanish.
3. $= n^{-2}(n\sigma^2) = \sigma^2/n$.

[UC Berkeley CS70 (2017), Note18: mean, variance, concentration]

Monte Carlo estimation

Independence makes the variance shrink

Convert variance to standard error

Standard error is the standard deviation of the estimator.

1. $\mathrm{SE}(\hat{\mu}_n) = \sqrt{\mathrm{Var}(\hat{\mu}_n)} = \sigma/\sqrt{n}$.
2. Four times as many independent samples gives half the standard error.
3. This describes the average, not the spread of one observation.

UC Berkeley CS70 (2017), Note18: mean, variance, concentration

EXERCISE

Monte Carlo estimation

What does four times as much data buy?

Independent draws have the same finite variance. Increase n from 100 to 400.

$$\mathrm{SE}(\hat{\mu}_n) = \sigma / \sqrt{n}$$

σ unchanged one-trial standard deviation

A Standard error becomes one quarter

B Standard error becomes one half

C Every realized estimate is closer

UC Berkeley CS70 (2017), Note18: mean, variance, concentration

DERIVATION

Monte Carlo estimation

A variance bound gives a probability statement

Chebyshev’s inequality bounds deviation probability using variance. Here $\hat{\mu}_n$ is an unbiased iid average; n is fixed and $\varepsilon>0$.

$$P(|\hat{\mu}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}$$

`epsilon` chosen error tolerance

`σ²` finite variance of g(X)

[UC Berkeley CS70 (2017), Note18: mean, variance, concentration]

Monte Carlo estimation

A variance bound gives a probability statement

Bound squared error on the bad event

$E=|\hat{\mu}_n-\mu|$, $\varepsilon>0$, and $A=\{E\geq\varepsilon\}$.

1. On A, $E^2\geq\varepsilon^2$; outside A, $E^2\geq 0$.
2. Therefore $\mathbb{E}[E^2] \geq \varepsilon^2 P(A)$.

[UC Berkeley CS70 (2017), Note18: mean, variance, concentration]

Monte Carlo estimation

A variance bound gives a probability statement

Insert the estimator variance

The fixed-n iid average is unbiased and has variance σ²/n.

1. $\mathbb{E}[E^2] = \mathrm{Var}(\hat{\mu}_n) = \sigma^2/n.$
2. Dividing gives $P(|\hat{\mu}_n - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2).$
3. The result is an upper bound, not an equality or a guarantee for each dataset.

UC Berkeley CS70 (2017), Note18: mean, variance, concentration

EXAMPLE

Monte Carlo estimation

A conservative trial count for a success rate

COURSE-CREATED WORKED EXAMPLE

Bernoulli success has variance $p(1-p) \le 1/4$. Target error 0.1 with failure probability ≤ 0.05.

$$\frac{1}{4n(0.1)^2} \le 0.05 \quad \Longrightarrow \quad n \ge 500$$

- p unknown success probability
- n fixed number of independent trials

[UC Berkeley CS70 (2017), Note18: mean, variance, concentration]

EXERCISE

Monte Carlo estimation

A tighter tolerance changes sample cost

Use p(1−p)≤1/4 and the variance bound. Keep failure probability ≤0.05; reduce error tolerance from 0.1 to 0.05.

$$P(|\hat{p} - p| \geq \varepsilon) \leq 1/(4n\varepsilon^2)$$

`p-hat,p` sample fraction and true success probability

A 250 trials suffice by this bound

B 1000 trials suffice by this bound

C 2000 trials suffice by this bound

UC Berkeley CS70 (2017), Note18: mean, variance, concentration

RESULT

Monte Carlo estimation

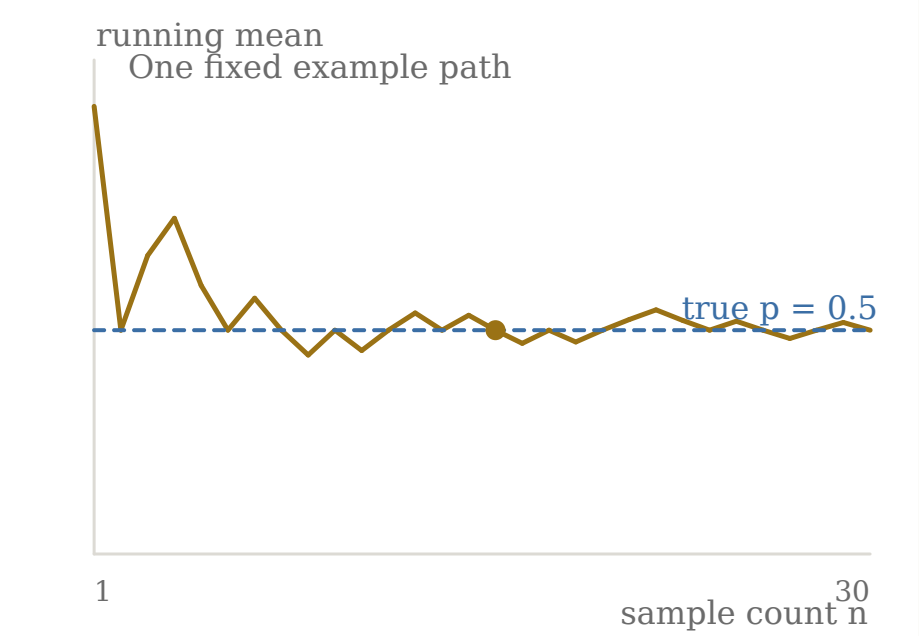
The law of large numbers is not a monotone path

Convergence in probability means every fixed positive error tolerance is exceeded with probability tending to zero. Here samples are iid with finite variance.

$$P(|\hat{\mu}_n - \mu| \geq \varepsilon) \longrightarrow 0$$

$\hat{\mu}_n$ average of n iid measurements

μ finite-variance population mean



UC Berkeley CS70 (2017), Note18: mean, variance, concentration

COUNTEREXAMPLE

Monte Carlo estimation

Correlated repetitions are not fresh evidence

Let $\bar{Z}$ be the mean of n measurements Z_i with finite variances. Dependence leaves pairwise covariance terms in its variance.

$$\operatorname{Var}(\bar{Z}) = \frac{1}{n^2}\left[\sum_i \operatorname{Var}(Z_i) + 2\sum_{i<j} \operatorname{Cov}(Z_i, Z_j)\right]$$

`Z_i` measured values g(X_i)

`Z-bar` sample average

UC Berkeley CS70 (2017), Note18: mean, variance, concentration

DEFINITION

Stopping rules

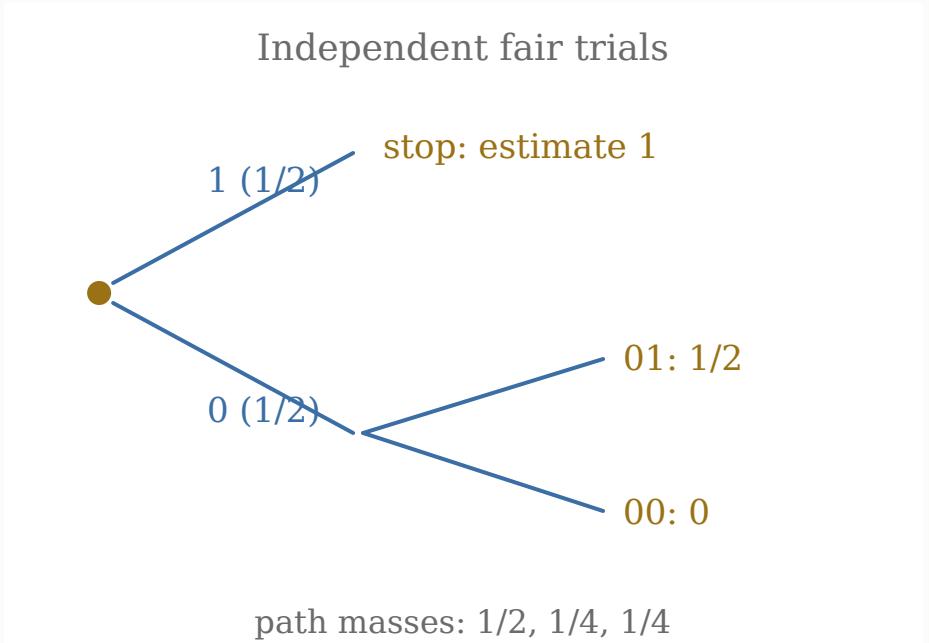
The data can determine when collection ends

A stopping time T is a random end time: whether $T \le t$ can be decided from observations $X_1,\dots,X_t$ alone.

$$T = \begin{cases} 1, & X_1 = 1 \\ 2, & X_1 = 0 \end{cases}$$

`X_i` independent fair Bernoulli trials

`T` random sample count chosen by the rule



Sheffield, MIT 18.175 (2016), Lecture 14, pp.34–35; original example

EXAMPLE

Stopping rules

Enumerating paths exposes the selection effect

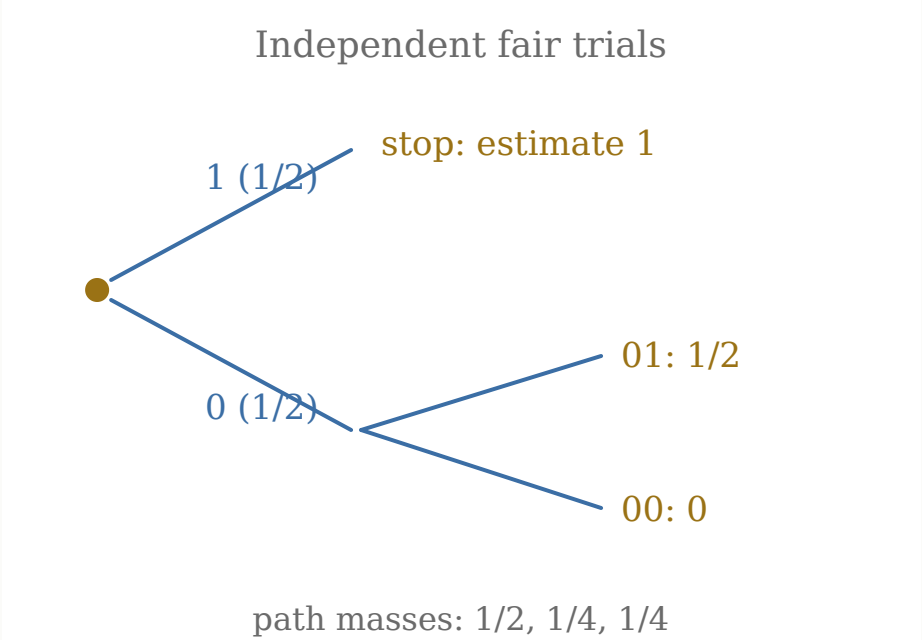
COURSE-CREATED WORKED EXAMPLE

Stop after first success, but take at most two trials. Each trial succeeds with probability 1/2.

$$(T, \hat{p}_T) = (1, 1), (2, 1/2), (2, 0)$$

`path probabilities` 1/2, 1/4, 1/4 respectively

`p̂T` success fraction at the stopping time



Sheffield, MIT 18.175 (2016), Lecture 14, pp.34–35; original example

EXERCISE

Stopping rules

Is the stopped success fraction unbiased?

Fair trials: p=1/2. Completed paths 1,01,00 have probabilities 1/2,1/4,1/4 and success fractions 1,1/2,0.

A Its expectation is 1/2

B Its expectation is 5/8

C Its expectation is 1

Sheffield, MIT 18.175 (2016), Lecture 14, pp.34–35; original example

EXERCISE

Stopping rules

Reverse the stopping preference

Trials are independent Bernoulli(1/2). Stop after the first failure or after two trials. Find the expected reported success fraction.

A. Expected estimate 3/8

B. Expected estimate 1/2

C. Expected estimate 5/8

Sheffield, MIT 18.175 (2016), Lecture 14, pp.34–35; original example

APPLICATION

Stopping rules

Match the guarantee to the collection rule

Fixed-n estimates need fixed-n evaluation. Adaptive collection needs an analysis designed for its stopping rule.

Howard et al. (2020), Introduction: time-uniform bounds

EXAMPLE

Monte Carlo estimation

A rollout result needs an uncertainty statement

COURSE-CREATED WORKED EXAMPLE

An independent fixed evaluation has 80 successes in 100 trials. Use a Bernoulli model.

$$\hat{p} = 0.8, \qquad \widehat{\mathrm{SE}} = \sqrt{\frac{0.8(0.2)}{100}} = 0.04$$

`p̂` observed success fraction

`estimated SE` plug-in standard error, not exact coverage

NIST Dataplot, Proportion Confidence Limits, Normal approximation

EXERCISE

Monte Carlo estimation

How many draws to see a rare route at least once?

A route has probability 0.1 in each independent draw. Find the smallest fixed integer n with detection probability at least 0.95.

$$P(\text{no rare route}) = (0.9)^n$$

n fixed independent sample count

A At least 10 draws

B At least 29 draws

C At least 95 draws

UC Berkeley CS70 (2017), Note18: mean, variance, concentration

MOTIVATION

Density and modes

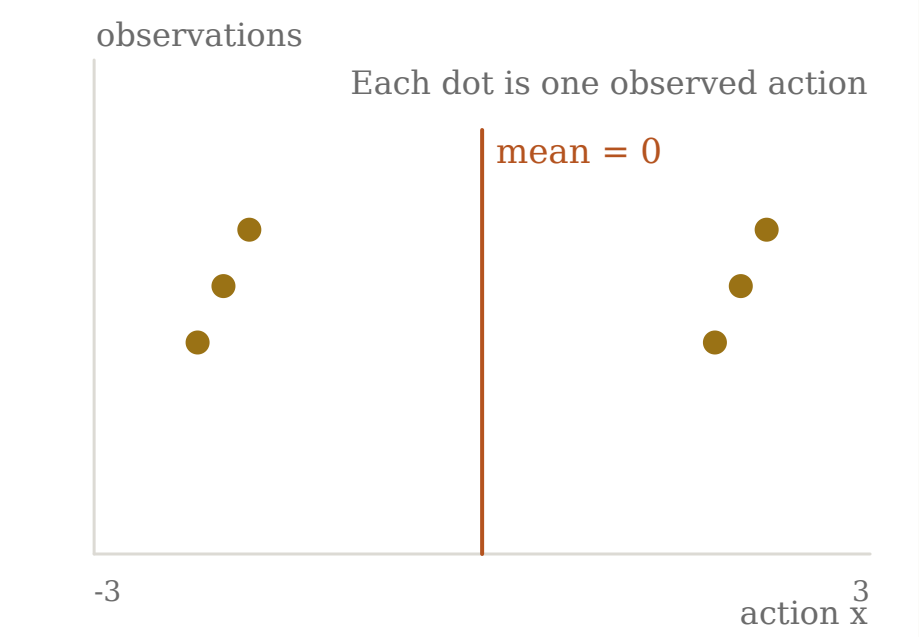
A mean can hide the choices a robot actually makes

Observed actions: -2.2, -2.0, -1.8, 1.8, 2.0, 2.2. Their mean is zero.

$$\bar{x} = 0, \qquad \text{dense groups lie near } -2 \text{ and } 2$$

`x_i` six observed scalar actions

`x̄` arithmetic mean



Wasserman, 36-708 Density Estimation, §§1,3–4

DEFINITION

Density and modes

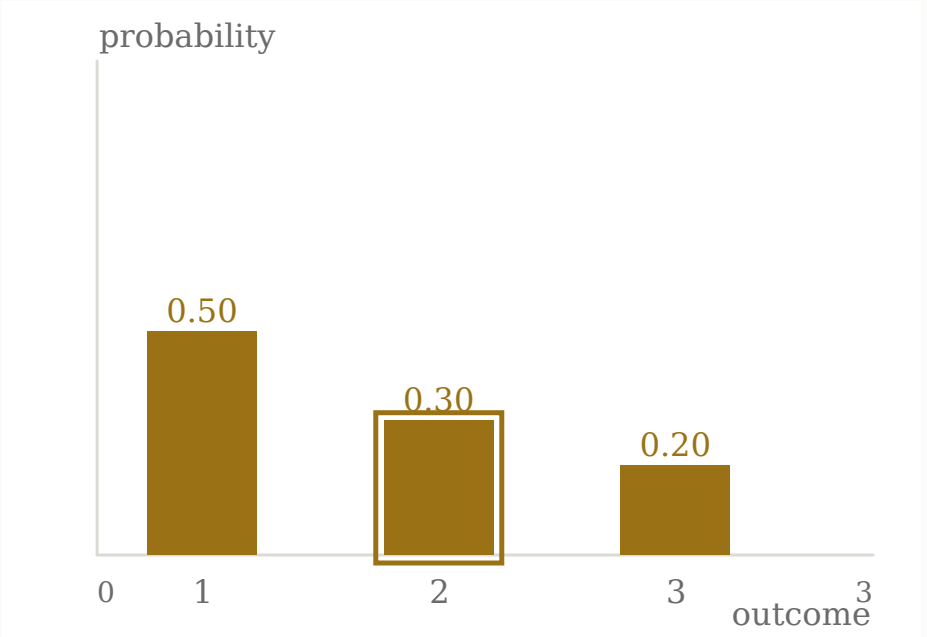
A discrete mode maximizes probability mass

A discrete mode maximizes true probability mass. An empirical mode maximizes observed frequency; tied maxima are possible.

$$\hat{p}_n(x) = \frac{N_x}{n}, \qquad \hat{m} \in \arg\max_x \hat{p}_n(x)$$

`N_x` number of observations equal to x

`m̂` one empirical mode; ties are possible



Wasserman, 36-708 Density Estimation, §§1,3–4

EXERCISE

Density and modes

Does the largest observed count prove the true mode?

Three categories have counts 5,3,2 in ten independent observations.

A. The first category is the empirical mode

B. The first category is certainly the population mode

C. The sample mean defines the mode

Wasserman, 36-708 Density Estimation, §§1,3–4

RECALL

Density and modes

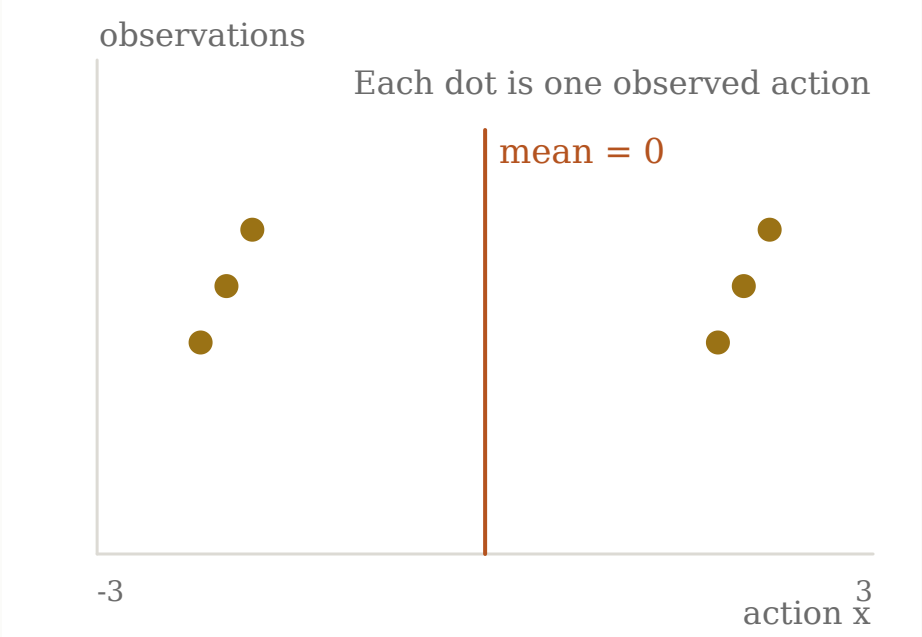
Continuous data rarely repeat exactly

For an absolutely continuous model, each exact value has probability zero, while intervals can have positive probability.

$$P(X = x) = 0, \qquad P(a \le X \le b) = \int_a^b p(u)\,du$$

`p(u)` density, not point probability

`a,b` interval endpoints



Goodfellow et al., Deep Learning (2016), §3.3.2

DEFINITION

Histograms

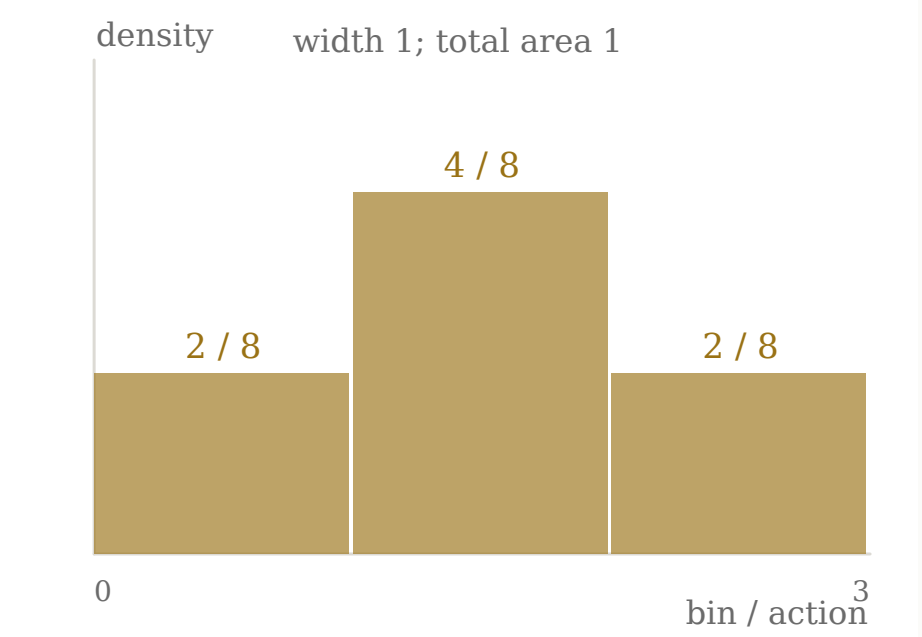
Counts become density after dividing by width

A histogram estimates density as a constant on each bin. Here bins have width h and contain counts N_j out of n observations.

$$\hat{p}(x) = \frac{N_j}{nh} \quad (x \text{ in bin } j)$$

`n,N_j` total count and bin count

`h` bin width in action units



[Wasserman, 36-708 Density Estimation, §3; original example]

EXAMPLE

Histograms

The area check catches normalization mistakes

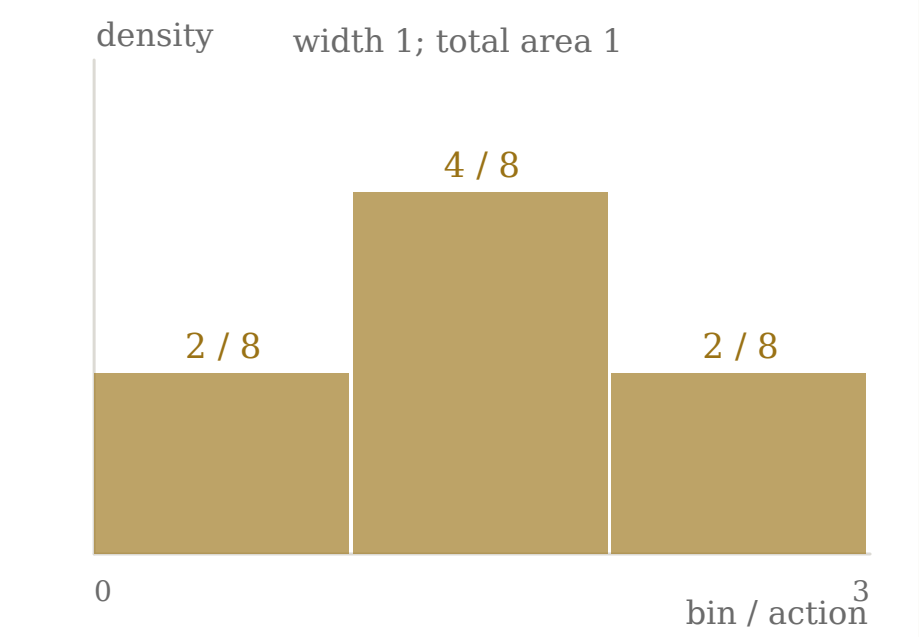
COURSE-CREATED WORKED EXAMPLE

Eight samples give counts 2,4,2 in three bins of width 1.

$$\hat{p} = (0.25, 0.5, 0.25), \qquad \sum_j h\hat{p}_j = 1$$

`h` common bin width 1

`p-hat_j` constant density in bin j



[Wasserman, 36-708 Density Estimation, §3; original example]

EXERCISE

Histograms

Compare bins with different widths

n=7. Bin A: count 4,width 2. Bin B: count 3,width 1. Which has higher histogram density?

$$\hat{p}_j = N_j / (n h_j)$$

`h_j` width of bin j; N_j is its count

A Bin A

B Bin B

C They have equal density

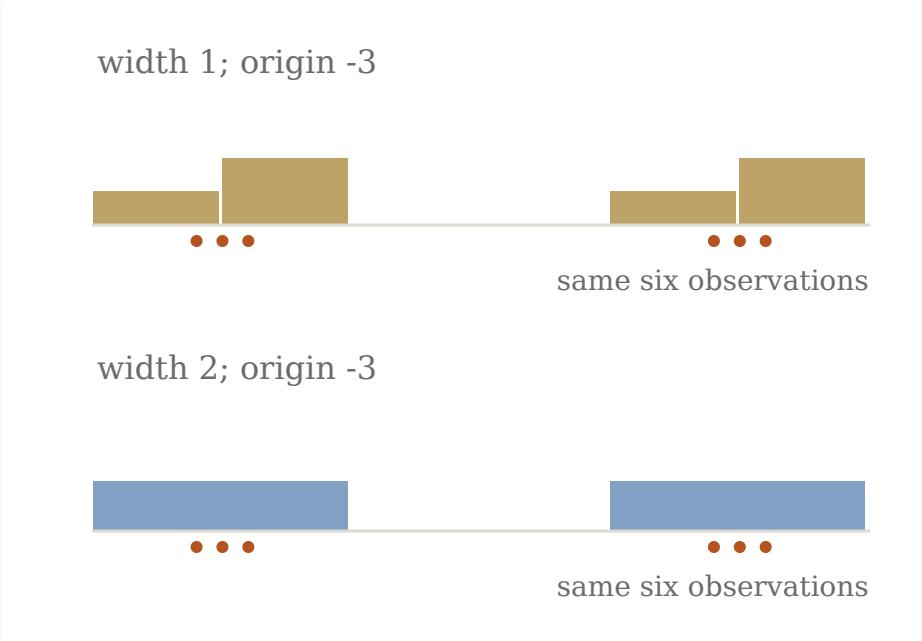
Wasserman, 36-708 Density Estimation, §3; original example

INTERPRETATION

Histograms

The apparent peak depends on the binning

Use observations $-2.2, -2, -1.8, 1.8, 2, 2.2$. Compare bin widths 1 and 2 with origin -3; changing bin boundaries also changes counts.



[Wasserman, 36-708 Density Estimation, §3; original example]

DEFINITION

Kernel density estimation

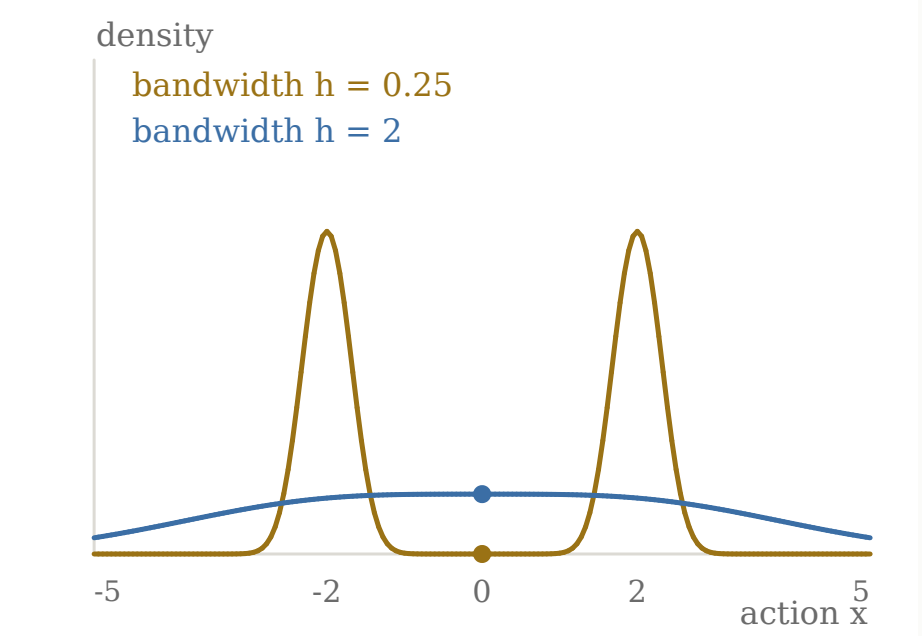
Each sample contributes a small smooth bump

Kernel density estimation (KDE) averages a rescaled bump at each sample. Use K≥0 with integral 1 and bandwidth h>0.

$$\hat{p}_h(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h} \right)$$

`x_i,x` observed samples and query location

`h,K` bandwidth and unit-integral kernel



[Wasserman, 36-708 Density Estimation, §4; original example]

DERIVATION

Kernel density estimation

The normalization survives a change of scale

Substitute $u=(x-x_i)/h$, so $dx=h\,du$.

$$\int \frac{1}{h} K\left(\frac{x - x_i}{h}\right) dx = \int K(u)\, du = 1$$

- `h` positive bandwidth
- `x_i` fixed sample center
- `u` rescaled integration variable

Wasserman, 36-708 Density Estimation, §4; original example

Kernel density estimation

The normalization survives a change of scale

Change the integration variable

K has integral 1, h>0, and x_i is a fixed sample center.

1. Set $u = (x - x_i)/h$, so $dx = h\,du$.
2. $\int h^{-1} K((x - x_i)/h)\,dx = \int h^{-1} K(u) h\,du = 1.$

Wasserman, 36-708 Density Estimation, §4; original example

Kernel density estimation

The normalization survives a change of scale

Average the unit-area bumps

There are n kernel bumps and the estimator divides their sum by n.

1. $\int \hat{p}_h(x)\,dx = n^{-1}\sum_{i=1}^{n} 1 = 1.$
2. Without the 1/h factor, each bump would instead have area h.
3. Bandwidth changes width and height together, preserving total probability.

Wasserman, 36-708 Density Estimation, §4; original example

EXERCISE

Kernel density estimation

What does a larger bandwidth change?

Use the same samples and a normalized Gaussian kernel. Increase h.

$$\hat{p}_h(x) = \frac{1}{nh} \sum_i K((x - x_i)/h)$$

`x,x_i` query location and fixed observations

`n,h` sample count and positive bandwidth

A Wider bumps, same total area

B More total probability mass

C Narrower bumps and more sharp peaks

Wasserman, 36-708 Density Estimation, §4; original example

EXAMPLE

Kernel density estimation

The same actions support different apparent shapes

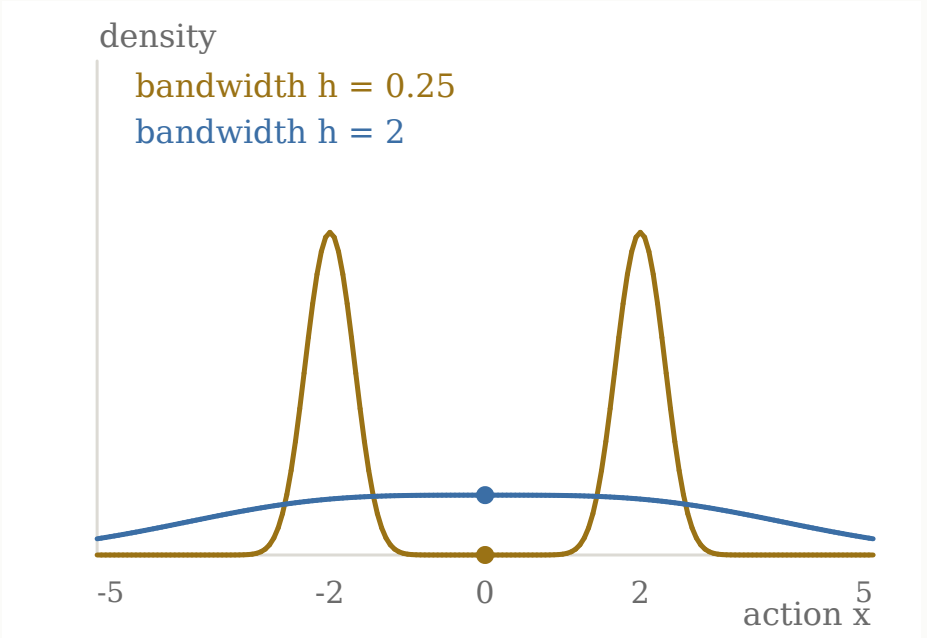
COURSE-CREATED WORKED EXAMPLE

A density mode maximizes density height. Estimate it from actions ±1.8,±2,±2.2 using Gaussian kernels with bandwidths 0.25 and 2.

$$\hat{m}_h \in \arg\max_x \hat{p}_h(x)$$

$\hat{m}_h$ estimated density mode, dependent on h

h bandwidth, in action units



Wasserman, 36-708 Density Estimation, §4; original example

EXERCISE

Density and modes

Choose an action from two observed groups

Actions form two separated groups around -2 and 2. A robot needs one plausible action.

A. Always execute the mean 0

B. Model the distribution and check the sampled action

C. Declare every observed value a true mode

Wasserman, 36-708 Density Estimation, §§1,3–4

DEFINITION

Stable probability computation

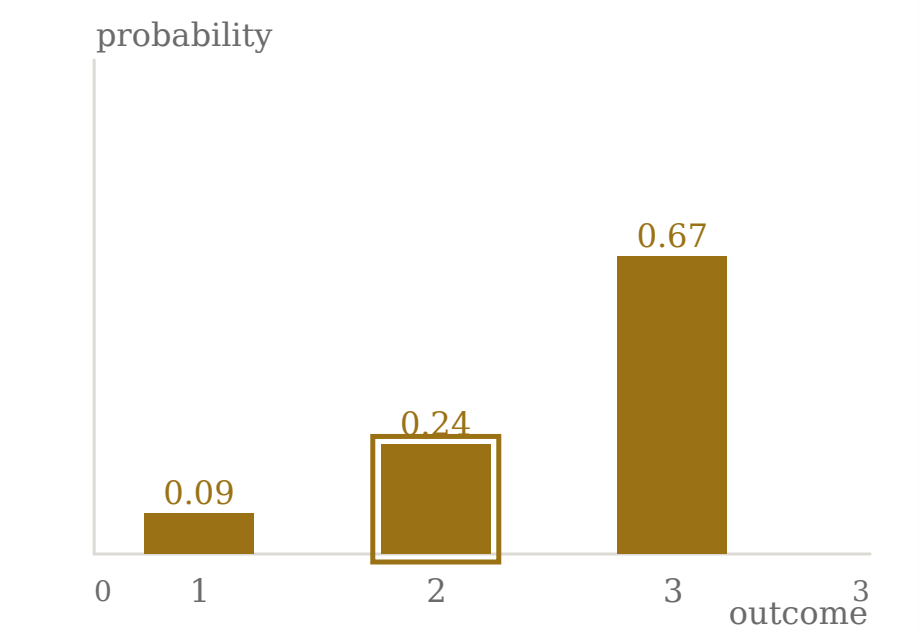
Exponentials turn scores into probabilities

Softmax converts finite real scores, called logits, into positive class probabilities that sum to one.

$$q_i = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}$$

`z_i` finite real logit for class i

`q_i,K` class probability and number of classes



Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

DERIVATION

Stable probability computation

A shared shift leaves softmax unchanged

Multiply numerator and denominator by e^{-c}. Choose $c=\max_j z_j$.

$$q_i = \frac{e^{z_i - c}}{\sum_j e^{z_j - c}}$$

`c` largest finite logit

`z_i−c` nonpositive shifted logit

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

Stable probability computation

A shared shift leaves softmax unchanged

Cancel a common exponential factor

z_i are finite logits; c is any finite shared constant.

1. $q_i = \frac{e^{z_i} e^{-c}}{\sum_j e^{z_j} e^{-c}}$.
2. $= \frac{e^{z_i - c}}{\sum_j e^{z_j - c}}$ by exponential multiplication.

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

Stable probability computation

A shared shift leaves softmax unchanged

Choose the shift for numerical safety

Set $c=\max_j z_j$.

1. Every shifted score satisfies $z_i-c\le 0$.
2. Therefore $0<\exp(z_i-c)\le 1$ in exact arithmetic; at least one equals 1.
3. This avoids exponential overflow. Tiny terms can still underflow in finite precision.

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

EXAMPLE

Stable probability computation

Large logits can encode ordinary probabilities

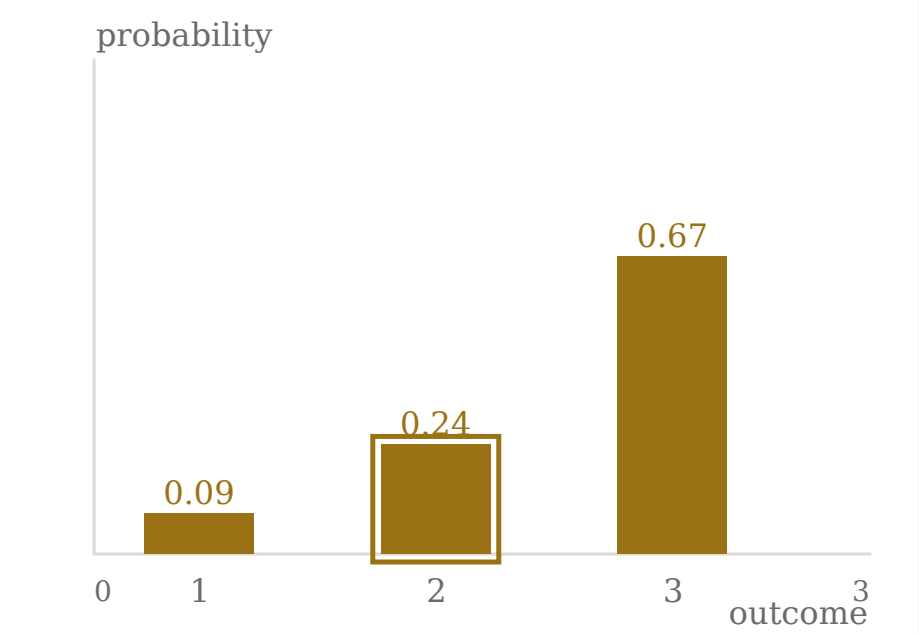
COURSE-CREATED WORKED EXAMPLE

Given z=(1000,1001,1002), subtract c=1002.

$$q = \frac{(e^{-2}, e^{-1}, 1)}{e^{-2} + e^{-1} + 1} \approx (0.090, 0.245, 0.665)$$

`z,q` logit vector and probability vector

`c` maximum logit 1002



Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

EXERCISE

Stable probability computation

Which change preserves the distribution?

For finite logits z, compare two transformations before softmax.

A $z \to z + 1000 \cdot 1$

B $z \to 2z$

C $z \to -z$

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

DERIVATION

Stable probability computation

Log-sum-exp keeps the normalization in log space

Log-sum-exp (LSE) is the natural logarithm of a sum of exponentials. Evaluate it using the largest finite logit c.

$$\mathrm{LSE}(z) = c + \log \sum_j e^{z_j - c}, \quad c = \max_j z_j$$

`LSE(z)` log of the sum of exponentials

`c` maximum finite logit

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

Stable probability computation

Log-sum-exp keeps the normalization in log space

Factor before taking the logarithm

$c = \max_j z_j$ for finite logits z.

1. $\sum_j e^{z_j} = e^c \sum_j e^{z_j - c}$.
2. $\log \sum_j e^{z_j} = \log e^c + \log \sum_j e^{z_j - c}$.
3. $\mathrm{LSE}(z) = c + \log \sum_j e^{z_j - c}$.

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

Stable probability computation

Log-sum-exp keeps the normalization in log space

Track the constant

LSE is a log normalization value, not a probability ratio.

1. Adding c to every logit adds c to LSE.
2. The shift term cancels in softmax ratios but must be restored in LSE.
3. For z=(1000,1000), LSE=1000+log 2.

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

DERIVATION

Stable probability computation

Compute log probabilities before exponentiating

For a known class y, use the log probability directly in cross entropy.

$$\log q_i = z_i - \mathrm{LSE}(z), \qquad \ell = -\log q_y$$

- y observed class index
- ℓ single-example negative log loss in nats

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

Stable probability computation

Compute log probabilities before exponentiating

Take the log of the softmax ratio

q_i is the class probability and $\mathrm{LSE}(z)=\log\Sigma_j\exp(z_j)$.

1. $\log q_i = \log e^{z_i} - \log \sum_j e^{z_j} = z_i - \mathrm{LSE}(z)$.
2. For the observed class y, $\ell = -\log q_y$.

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

Stable probability computation

Compute log probabilities before exponentiating

Keep the computation in shifted form

$c = \max_j z_j$ and y is the observed class index.

1. $\log q_i = (z_i - c) - \log \sum_j e^{z_j - c}$.
2. $\ell = \log \sum_j e^{z_j - c} - (z_y - c)$.
3. Average ℓ across observations to obtain empirical cross entropy in nats.

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

EXERCISE

Stable probability computation

Compute the observed-class log loss

z=(1000,999). The observed class is the second. Compute loss in nats.

$$\ell = \log \sum_j e^{z_j - c} - (z_y - c), \quad c = \max_j z_j$$

y observed class index; log is natural

A About 0.313

B About 1.313

C Infinite because e^{1000} overflows

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

DERIVATION

Stable probability computation

The gradient of log-sum-exp is softmax

Let $z, q \in \mathbb{R}^K$ be logit and probability columns. The one-hot target e_y has a 1 at class y and 0 elsewhere.

$$\frac{\partial}{\partial z_i} \log \sum_j e^{z_j} = \frac{e^{z_i}}{\sum_j e^{z_j}} = q_i$$

`z`	column vector of logits
`gradient`	column probability vector q

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

Stable probability computation

The gradient of log-sum-exp is softmax

Differentiate the scalar normalizer

$S=\Sigma_j \exp(z_j)$, and the logits are independent coordinates.

1. $\partial S/\partial z_i = e^{z_i}$.
2. $\partial \log S/\partial z_i = S^{-1}\partial S/\partial z_i = e^{z_i}/S = q_i$.

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

Stable probability computation

The gradient of log-sum-exp is softmax

Differentiate the observed-class loss

ℓ=LSE(z)−z_y; e_y is the one-hot column target.

1. $\nabla_z \operatorname{LSE}(z) = q$.
2. $\nabla_z z_y = e_y$, so $\nabla_z \ell = q - e_y$.
3. The gradient components sum to 1−1=0, consistent with shared-shift invariance.

Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

EXERCISE

Stable probability computation

The loss gradient changes the right scores

q=(0.7,0.2,0.1). The true class is 2. Use gradient descent on logits.

$$\nabla_z \ell = q - e_2, \qquad z_{new} = z - \eta \nabla_z \ell$$

`e_2` one-hot target (0,1,0)

`η` positive gradient-descent step size

A. Increase all logits equally

B. Increase class 2, decrease classes 1 and 3

C. Decrease class 2, increase classes 1 and 3

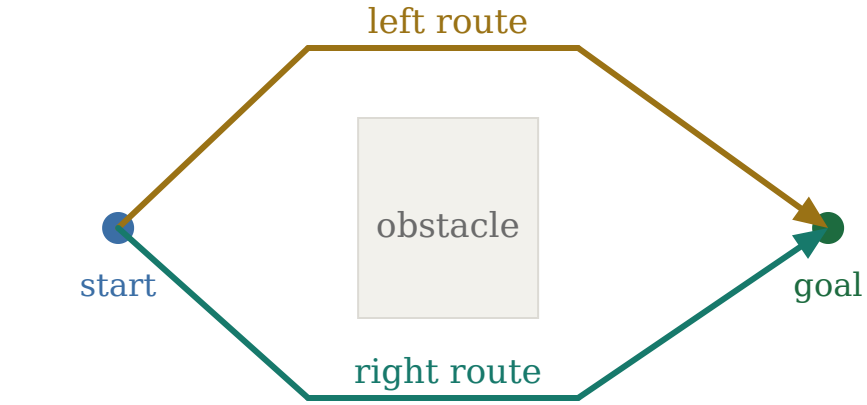
Blanchard, Higham & Higham, arXiv:1909.03469, §§1,4

SETUP

Robot synthesis

Choose the quantities before calculating them

Source p=(0.8,0.2), model q=(0.6,0.4). Plan 100 independent trials. Find H(p), H(p,q), KL, and the route-fraction standard error.



Goodfellow et al., Deep Learning (2016), §3.13

SOLUTION

Robot synthesis

Information and sampling answer different parts

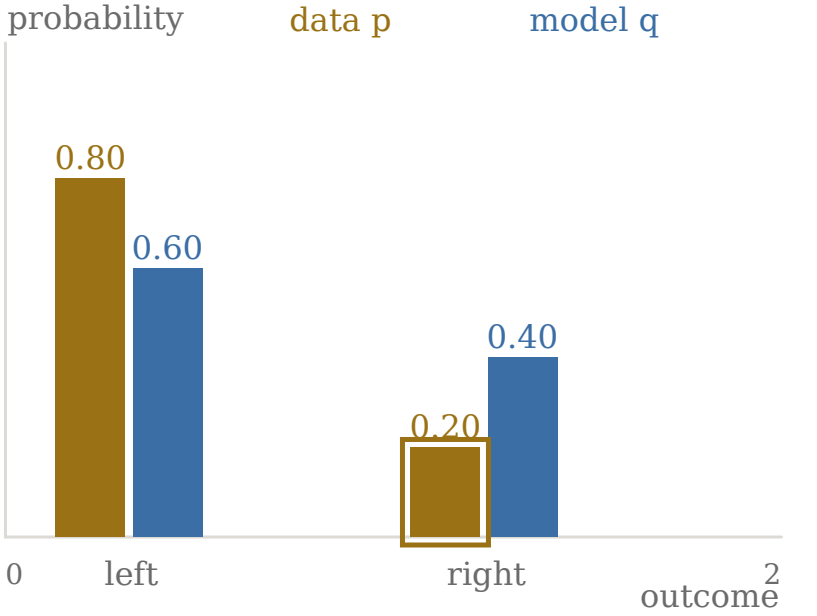
COURSE-CREATED WORKED EXAMPLE

p=(0.8,0.2), q=(0.6,0.4), n=100 independent trials.

$$H(p) \approx 0.722, \quad H(p,q) \approx 0.854, \quad D_{\mathrm{KL}} \approx 0.132$$

values bits per route; exact source probabilities given

SE sqrt(0.8·0.2/100)=0.04 for route frequency



Goodfellow et al., Deep Learning (2016), §3.13

EXERCISE

Robot synthesis

Evaluate a changed route frequency

Source changes to p=(0.2,0.8); model stays q=(0.6,0.4). Collect a fresh fixed evaluation.

A. Entropy stays the same; cross entropy rises

B. Both quantities stay the same

C. Entropy must increase

Goodfellow et al., Deep Learning (2016), §3.13

RECAP

The reasoning path

From scoring predictions to generating actions

Probabilities support log losses. Samples support estimates. Structured density models support multiple action choices.

Questions for the same robot

1. Compare route predictions
2. Measure sensor information
3. Assess evidence from trials
4. Locate action groups
5. Compute probabilities reliably

Goodfellow et al., Deep Learning (2016), §3.13

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Probability and information

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